PETROCHEMICALS: REVIEW 25

The petrochemical industry is of crucial importance in modern life. It supplies a large portion of our energy needs, such as petrol, diesel fuel and methane. From crude oil (petroleum) it supplies the raw materials used in the manufacture of many consumer products. Crude oil is a complex mixture of hydrocarbons.

1 What is a hydrocarbon?

Crude oil is of little use in its raw form. It has to be treated before it is of commercial use. The first step is fractional distillation in a refinery. This separates it into various fractions.

2 What difference in physical properties is utilised in fractional distillation?

Each fraction contains a mixture of hydrocarbons with similar chain length.

3 What is meant by *similar chain length*?

The proportion of hydrocarbons present in crude oil does not correspond with society's demands. The chemicals present are modified to match supply with demand. The long chains of alkanes present can be broken to form shorter (and more useful) compounds.

4 What is an *alkane*?

Cracking When alkanes are heated, their molecules vibrate. At high temperatures they vibrate so strongly that carbon–carbon bonds are broken and smaller molecules are formed. This process is called **thermal cracking**. One of these smaller molecules is an alkene.

5 What is an *alkene*?

By using a catalyst, thermal cracking can be made to occur at lower temperatures in a process called **catalytic cracking**.

6 Define *catalyst*.

This is the cracking of a typical alkane, undecane to nonane and ethene (ethylene, an alkene):

$$C_{11}H_{24}\ (l) \rightarrow C_9H_{20}\ (l) + C_2H_4\ (g)$$

7 Draw the structural formula for ethene.

8 What is the major structural difference between ethene and the two alkanes, undecane and nonane?

Because ethene has a double bond (**unsaturated**), it is a reactive compound. Ethene is the raw material in the formation of ethanol (C_2H_5OH), ethan-1, 2-diol (ethylene glycol, $HOCH_2CH_2OH$) and polythene.

9 Give one use for each compound.

- a ethanol
- b ethan-1, 2-diol
- c polythene

10 One way of forming ethanol is by reacting ethene with water using a catalyst at high temperature and pressure.

- a Write a balanced equation for this reaction.
- b Why do you think a catalyst was used?
- c Give the name of another process that produces ethanol.

11 Why is petroleum cracked?

12 Why is an alkene usually more reactive than the equivalent alkane?

13 Name three petrochemical fuels.

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